

Math 135

Honor Algebra

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1 Admin

1.1 General Information

1.1.1 Contact

For mathematics questions: `abeltaos@uwaterloo.ca`

For course logistics: `math135@uwaterloo.ca`

1.1.2 Teaching Style

The LEARN website contains several useful resources. In particular:

- (i) The icons at the top of the page provide quick access to important resources, including the course schedule.
- (ii) Crowdmark is used for submitting assignments.

1.1.3 Graded Assignments

There are three assignments each week.

1.1.4 Late Submission Policy

Late submissions are accepted with a penalty of 1% per minute for the entire assignment.

2 Introduction to the Language of Mathematics

2.1 Introduction to Proof

Definition 2.1 (Proof). A proof is a logically valid argument that establishes the truth of a mathematical statement using definitions, axioms, and previously proved results.

Proposition 2.2. *For every positive integer n , the number $n^2 + 1$ is not a perfect square.*

Proof. Let n be a positive integer. The consecutive perfect squares surrounding $n^2 + 1$ are n^2 and $(n + 1)^2$. Indeed,

$$n^2 < n^2 + 1 < n^2 + 2n + 1 = (n + 1)^2,$$

where the second inequality follows from $n > 0$.

Therefore, $n^2 + 1$ lies strictly between two consecutive perfect squares, so it cannot itself be a perfect square. ■